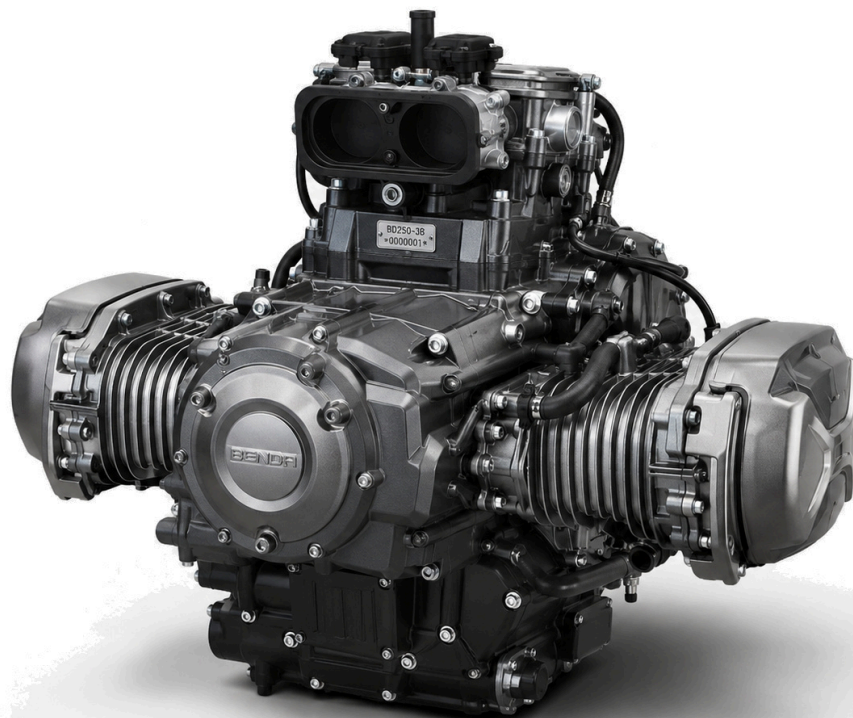


ZHEJIANG CHANGLING BENJIAN MOTORCYCLE CO., LTD.

浙江长铃奔健机车有限公司

ENGINE PRODUCT PRESENTATION

发动机产品说明



CB250B

250CC HORIZONTALLY OPPOSED TWIN-CYLINDER ENGINE

250cc 水平对置双缸发动机

FOUR-STROKE · LIQUID-COOLED
DOHC · 4 VALVES PER CYLINDER · EFI
四冲程 · 液冷 · 双顶置凸轮轴 · 电喷



250CC HORIZONTALLY OPPOSED TWIN-CYLINDER ENGINE, FRONT THREE-QUARTER VIEW

250cc 水平对置双缸发动机 · 前三分之四视图

A compact 250cc boxer powertrain platform

紧凑型 250cc 水平对置动力平台

CB250B 是一款面向现代摩托车动力系统开发的 250cc 级水平对置双缸发动机。发动机采用 180° 水平对置结构，两侧气缸沿曲轴中心线水平布置，在保持紧凑排量的同时形成低重心、结构对称、辨识度突出的动力平台。

The engine combines a four-stroke liquid-cooled boxer layout with DOHC four-valve cylinder heads and electronic fuel injection, and is configured for chain final drive in motorcycle applications.

AT A GLANCE 核心参数

Engine Model 发动机型号	CB250B
Configuration 发动机类型	180° opposed twin
Displacement 实际排量	249cc class △
Valvetrain 配气机构	DOHC, 8 valves
Cooling 冷却方式	Liquid-cooled
Fuel System 供油方式	Electronic fuel injection
Dry Weight 干重	approx. 48 kg △

△ Proposed engineering value pending factory confirmation. 建议工程值，待工厂确认。

01

Horizontally Opposed Twin

水平对置双缸

180° opposed cylinders on the crankshaft centreline.

02

Liquid Cooling

液冷系统

Active thermal management of barrels and heads.

03

DOHC Four-Valve

DOHC 四气门

Four camshafts, eight valves, high-rpm breathing.

04

Compact 250cc Platform

紧凑 250cc 平台

Low build height for lightweight chassis packaging.

03 ENGINEERING CONCEPT

结构概念

FRONT VIEW

正视图

A1
CYLINDER HEAD
气缸盖

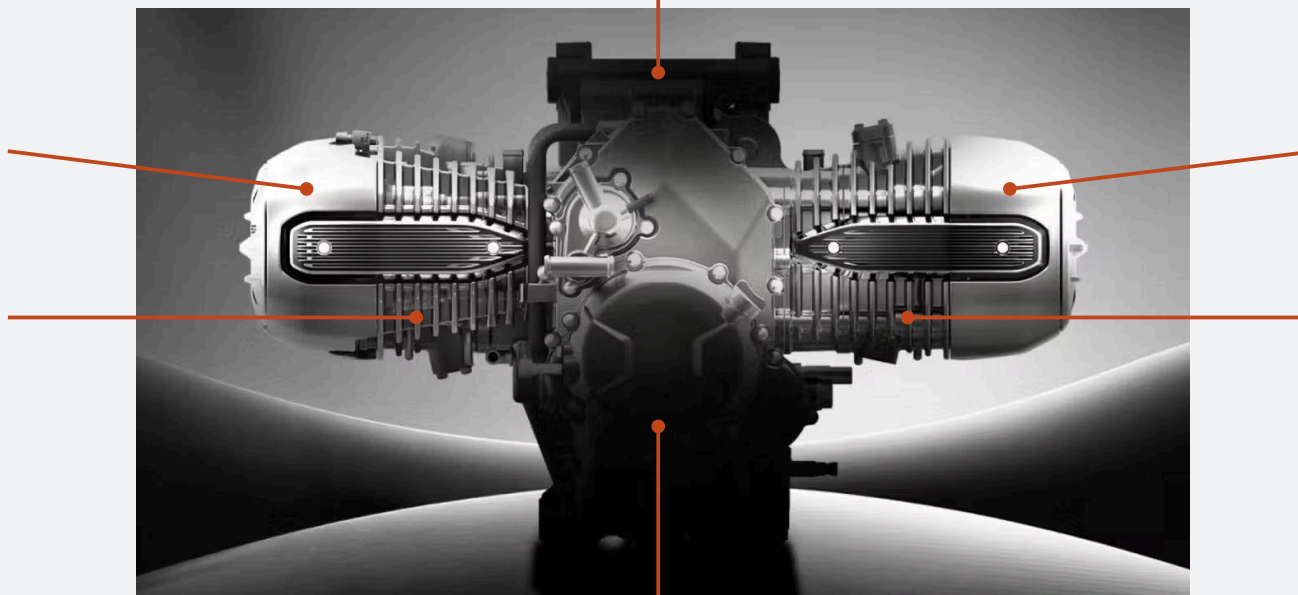
A2
LEFT CYLINDER
左侧气缸

A5
INTAKE
进气

A3
CYLINDER HEAD
气缸盖

A4
RIGHT CYLINDER
右侧气缸

A6
CRANKCASE
曲轴箱



Cylinders are arranged horizontally on either side of the crankcase, giving the engine lateral width and low vertical height.

进气位于发动机上部，排气位于下部，两侧气缸沿曲轴中心线对称布置。

Annotations identify externally visible assemblies only. No internal cross-section is represented.

04

KEY SPECIFICATIONS

主要技术参数

ENGINE MODEL

CB250B

ITEM 项目	CB250B
Engine Model 发动机型号	CB250B
Engine Type 发动机类型	Horizontally opposed twin
Cylinder Arrangement 气缸排列	180° opposed
Number of Cylinders 气缸数量	2
Operating Cycle 工作方式	Four-stroke
Displacement Class 排量级别	250cc
Actual Displacement 实际排量	249cc class △
Bore 缸径	53.5 mm △
Ignition 点火方式	ECU-controlled electronic
Fuel Type 燃料类型	Unleaded gasoline
Recommended Fuel 推荐燃油	RON 92 and above △

ITEM 项目	CB250B
Stroke 行程	55.2 mm △
Compression Ratio 压缩比	11.5 : 1 ▲
Valvetrain 配气机构	DOHC
Valves 气门数量	4 per cylinder / 8 total
Cooling 冷却方式	Liquid-cooled
Induction 进气方式	Naturally aspirated
Fuel System 供油方式	Electronic fuel injection
Starting 启动方式	Electric start
Final Drive Adaptation 最终传动适配	Chain drive
Transmission Interface 传动适配	Motorcycle gearbox
Dry Weight 干重	approx. 48 kg △

▲ Publicly reported figure — reported by technical media, not a factory-certified specification. 公开报道数据，非工厂认证参数。

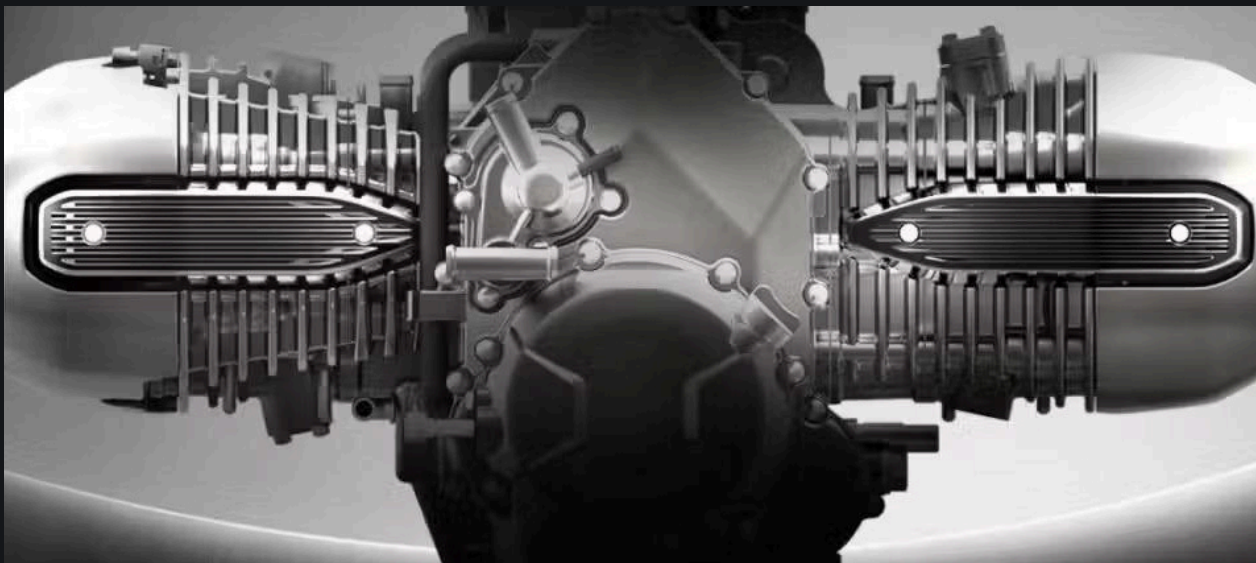
△ Proposed engineering value — supplied for catalog completion pending factory confirmation. 建议工程值，最终以生产图纸及工厂确认数据为准。

05 BOXER ARCHITECTURE

水平对置结构

180° OPPOSED

180° 对置



SYMMETRICAL OPPOSED CYLINDER ARRANGEMENT ABOUT THE CRANKSHAFT CENTRELINE

以曲轴中心线对称布置的水平对置气缸

01

Low Centre of Gravity

低重心

Principal engine mass sits low in the chassis, lowering powertrain centre of gravity.

03

Compact Platform

紧凑动力平台

250cc displacement in a boxer layout for lightweight motorcycle applications.

02

Symmetrical Layout

对称布局

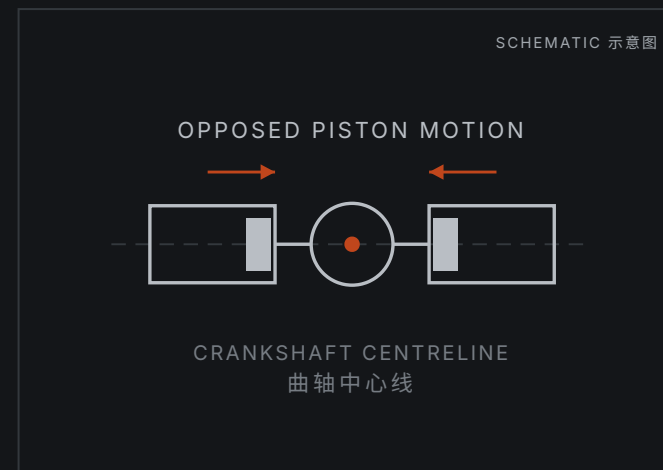
Left and right cylinders are mirrored, giving distinctive mechanical balance characteristics.

04

Distinctive Character

独特机械特性

Motion and power delivery differ from parallel-twin and V-twin configurations.



CONCEPTUAL REPRESENTATION OF THE 180° OPPOSED LAYOUT. NOT AN ENGINEERING DRAWING.

180° 水平对置布置概念示意，非工程图纸。

The two pistons travel in opposite directions along a common horizontal axis. The resulting mechanical character — and the lateral, low-profile package it creates — is the defining attribute of the CB250B platform.

水平对置结构使两个活塞呈相反方向运动，形成横向展开、纵向高度较低的动力结构。

06

DOHC FOUR-VALVE SYSTEM

DOHC 四气门系统

8 VALVES · 4 CAMSHAFTS

8 气门 · 4 凸轮轴



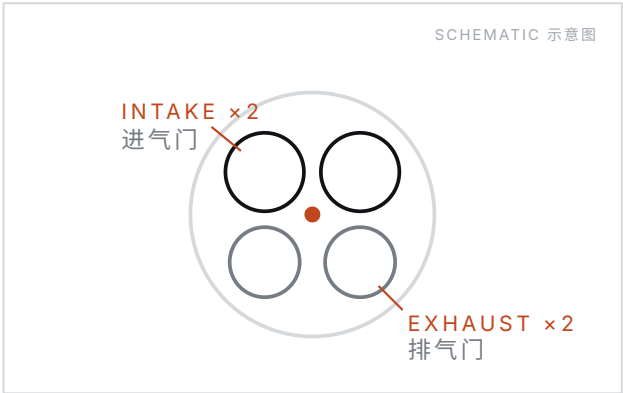
CYLINDER HEAD AND VALVE COVER ASSEMBLY, LEFT CYLINDER

左侧气缸盖及气门室盖组件

Double overhead camshafts operate four valves per cylinder. The increased effective port area improves gas exchange in the mid and upper rev range.

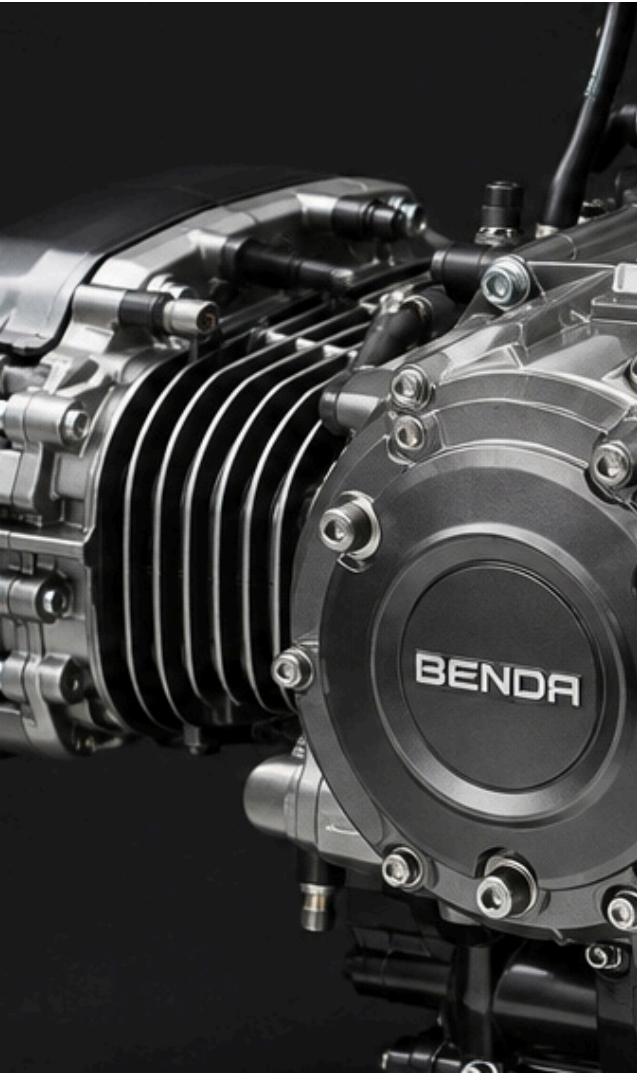
每缸四气门设计通过增加进排气有效面积，提高发动机在中高转速区域的换气效率，并为 250cc 级发动机提供更高的单位排量功率潜力。

VALVETRAIN 配气系统	VALUE
Valvetrain Type 配气形式	DOHC
Valves per Cylinder 每缸气门	4
Total Valves 总气门数量	8
Camshafts 凸轮轴数量	4
Intake Valves 进气门	2 / cylinder
Exhaust Valves 排气门	2 / cylinder



FOUR-VALVE ARRANGEMENT, CONCEPTUAL

四气门布置概念示意



CYLINDER BARREL, HEAD INTERFACE AND COOLANT HOSE ROUTING

气缸体、缸盖接口及冷却液管路走向

Stable thermal management

稳定热管理

A liquid cooling circuit provides active thermal management of the cylinder barrels, cylinder heads and high-temperature zones, holding operating temperature within a controlled band across load and ambient conditions.

冷却系统对缸体、缸盖和高温区域进行主动热管理，使发动机在不同负荷及环境条件下保持稳定的工作温度。

SYSTEM COMPONENTS 冷却系统组成

- Aluminium radiator 铝制散热器

— Electric cooling fan 电动冷却风扇

— Water pump 水泵

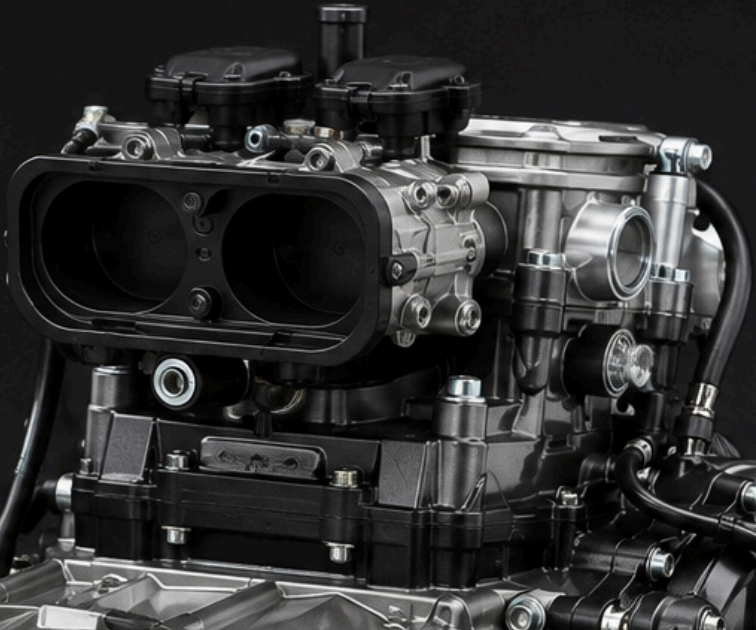
— Thermostat 节温器
- Coolant circulation passages 冷却液循环通道

— Cylinder water jackets 缸体冷却水道

— Cylinder head water jackets 缸盖冷却水道

PROPOSED 建议参数	VALUE
Coolant Capacity 冷却液容量	approx. 1.6 L [△]
Thermostat Opening 节温器开启	82 °C [△]
Fan Activation 风扇启动	100 °C [△]
Coolant Type 冷却液类型	Long-life ethylene glycol

[△] Proposed engineering value pending factory confirmation. 建议工程值，待工厂确认。Coolant temperatures are not factory-certified data. 冷却温度非工厂认证数据。



UPPER INTAKE ASSEMBLY — AIRBOX INTERFACE, THROTTLE BODIES AND INJECTOR MOUNTING

上部进气组件 · 进气箱接口、节气门体与喷油器安装位



FUEL SYSTEM 供油系统	VALUE
Fuel System 供油方式	EFI
Throttle Body 节气门体	34 mm [△]
Injectors 电子喷油器	2 [△]
Fuel Pressure 燃油压力	3.0 bar [△]
Oxygen Sensor 氧传感器	Wideband [△]

IGNITION 点火系统	VALUE
Ignition Type 点火方式	ECU electronic
Ignition Coils 点火线圈	2 [△]
Spark Plugs 火花塞	2 [△]
Plug Gap 火花塞间隙	0.7–0.8 mm [△]

[△] Proposed engineering value. 建议工程值，待工厂确认。

ECU CONTROL FUNCTIONS ECU 控制功能

- Fuel metering 燃油喷射量

— Ignition timing 点火提前角

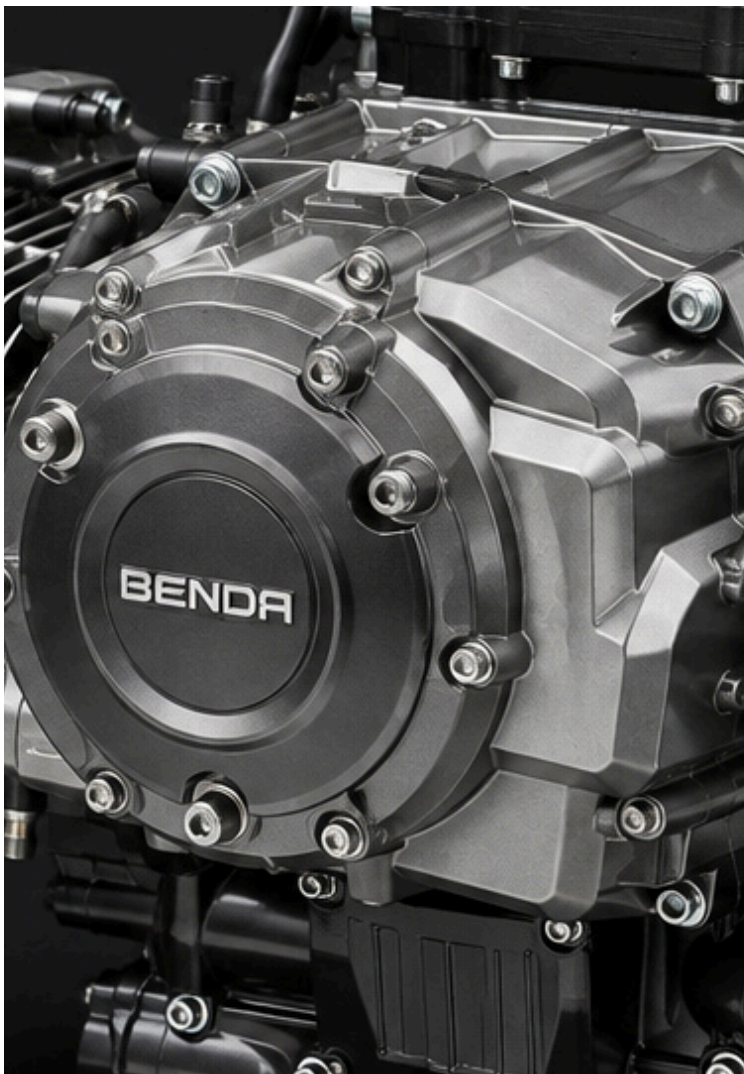
— Idle control 怠速控制
- Temperature compensation 温度补偿

— Engine protection 发动机保护

— Per-cylinder control 双缸独立控制

ECU inputs — engine speed · throttle position · intake pressure · intake temperature · coolant temperature · oxygen sensor feedback.

ECU 输入信号 — 发动机转速 · 节气门开度 · 进气压力 · 进气温度 · 冷却液温度 · 氧传感器反馈。



180° boxer crankshaft layout

180° 水平对置曲轴结构

The crankshaft layout is matched to the horizontally opposed cylinder arrangement, with the two pistons moving in opposite directions along a shared horizontal axis.

曲轴采用高强度合金钢制造，并经过精密加工、动平衡及热处理，以满足高转速工况下的强度与平衡要求。

ASSEMBLY FUNCTIONS 曲轴系统功能

- Reciprocating-to-rotary conversion 往复运动转换
- Power output 动力输出
- Crank phase control 曲柄相位控制
- Twin-cylinder motion coordination 双缸运动协调
- Vibration management 振动管理
- Connecting rod support 连杆支承

MATERIALS & PROCESS 材料与工艺

- High-strength alloy steel 高强度合金钢
- Precision machining 精密加工
- Dynamic balancing 动平衡
- Heat treatment 热处理

Internal components are described functionally. No internal geometry is illustrated in this document. 内部构件仅作功能性描述，本文件不表现内部几何结构。

CENTRAL CRANKCASE AND SIDE COVER — CRANKSHAFT AND
TRANSMISSION INTERFACE HOUSING

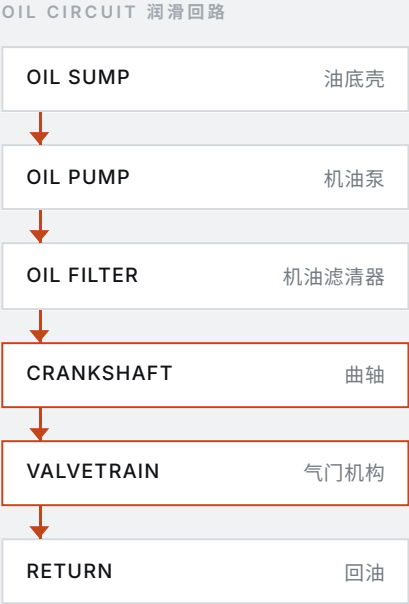
中央曲轴箱与侧盖 · 曲轴及传动机构安装壳体



OIL FILTER AND LOWER CRANKCASE / SUMP AREA
机油滤清器及下曲轴箱、油底壳区域

LUBRICATED ASSEMBLIES 润滑区域

- Crankshaft & connecting rods 曲轴与连杆
- Pistons 活塞
- Camshafts & valvetrain 凸轮轴与气门机构
- Main bearings 主轴承
- Transmission gears 传动机构



CONCEPTUAL CIRCUIT SEQUENCE.
ACTUAL OIL ROUTING PER PRODUCTION
DRAWINGS.
概念性回路顺序，实际油路以生产图纸为准。

Full pressure lubrication

全机强制润滑

A pressure lubrication system supplies continuous oil flow to the principal moving components of the engine, supporting sustained high-rpm operation.

压力润滑系统为发动机关键运动部件提供持续润滑，保证高转速工况下的可靠油膜供应。

PROPOSED 建议参数	VALUE
Lubrication Type 润滑方式	Wet sump, pressure-fed
Oil Capacity 机油容量	1.8 L [△]
Refill Capacity 换油容量	approx. 1.6 L [△]
Recommended Oil 推荐机油	10W-40 full synthetic [△]
Oil Grade 机油等级	API SN / JASO MA2 [△]

[△] Proposed engineering value pending factory confirmation. 建议工程值，待工厂确认。

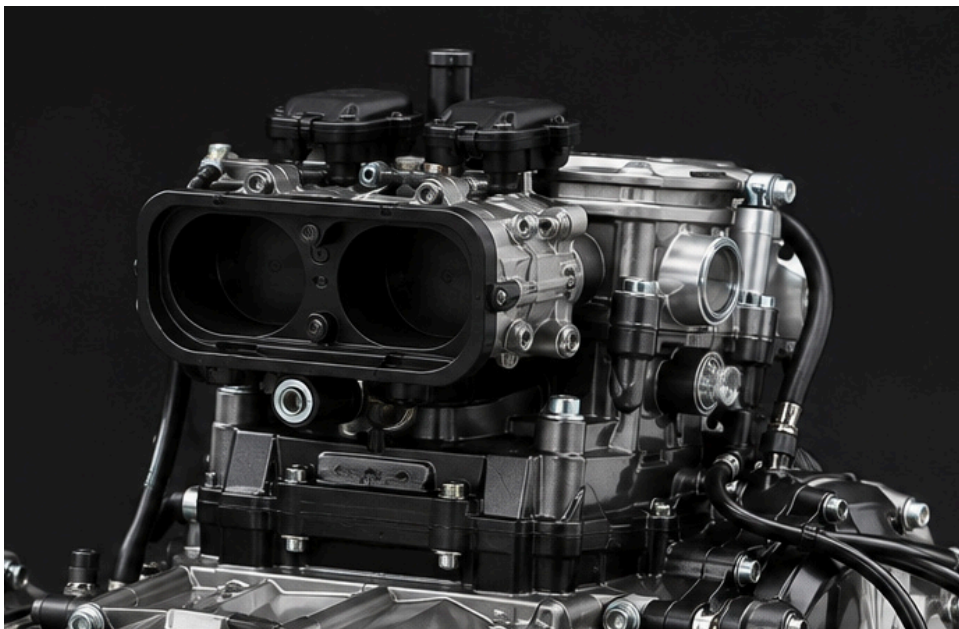
11 INTAKE & EXHAUST

进气与排气

UPPER INTAKE · LOWER EXHAUST

上进气 · 下排气

INTAKE ↑ UPPER INTAKE 上部进气



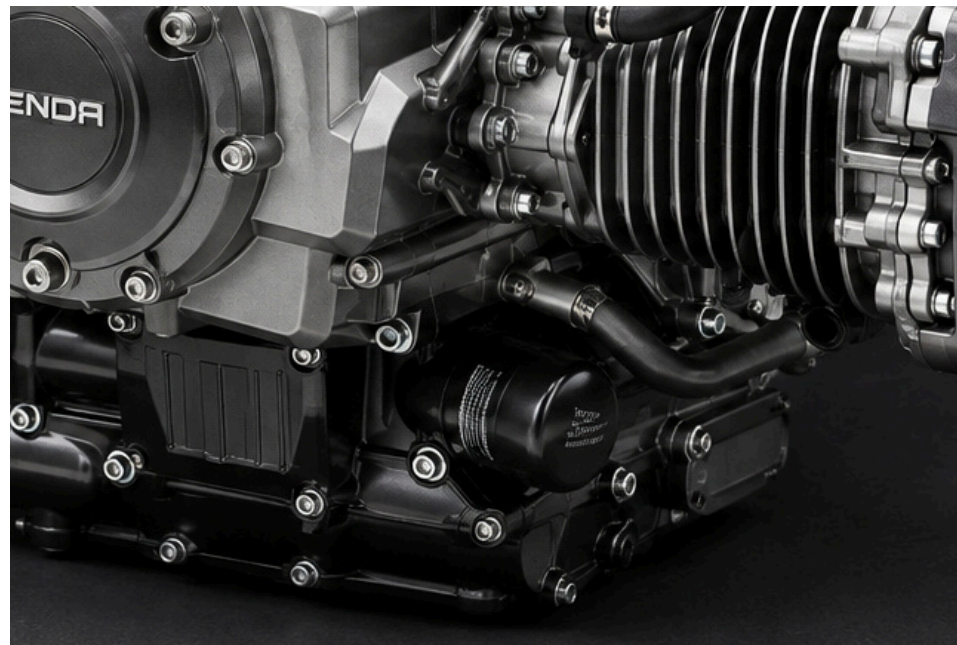
Fresh air is fed to both cylinders through a short, direct path from the airbox and throttle bodies mounted above the crankcase.

进气系统通过短而直接的进气路径向两个气缸提供新鲜空气。

COMPONENTS 主要部件

- Air filter 空气滤清器
- Airbox 进气箱
- Throttle body 节气门体
- Intake manifold 进气歧管
- Injectors 喷油器
- Intake valves 进气门

EXHAUST ↓ LOWER EXHAUST 下部排气



Exhaust gas is discharged from the lower section of the engine. Final pipe routing and silencer specification are matched to the vehicle platform and applicable emissions regulations.

排气系统应根据最终车型的排放法规及整车空间进行匹配。

COMPONENTS 主要组成

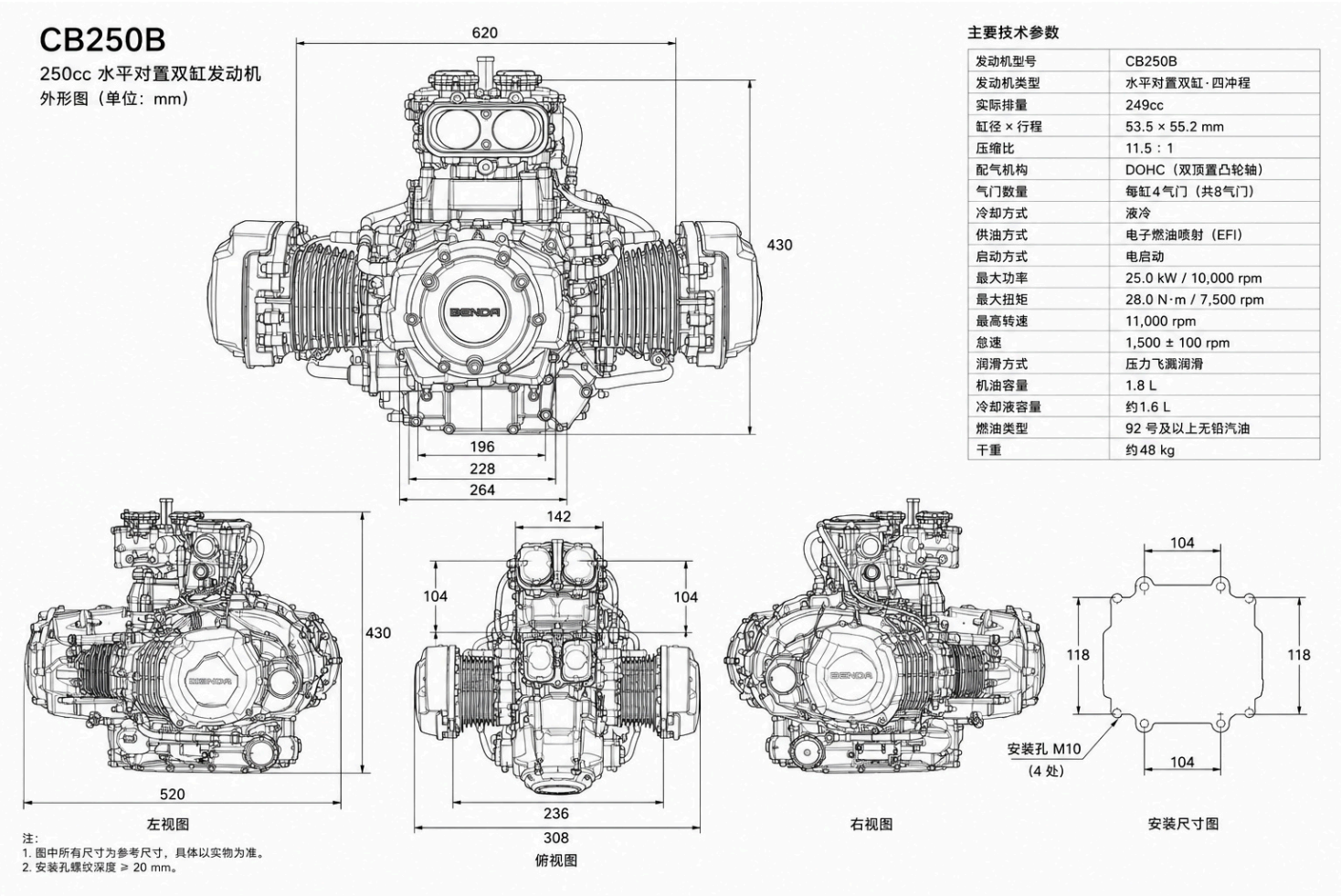
- Exhaust valves 排气门
- Exhaust manifold 排气歧管
- Three-way catalyst 三元催化器
- Silencer 消声器
- Oxygen sensor 氧传感器

Layout direction only. Exact pipe routing is defined by the vehicle installation and production drawings. 仅表示布置方向，具体管路走向由整车安装及生产图纸确定。

12 DIMENSIONS & MOUNTING INTERFACE

外形尺寸与安装接口

UNIT: MM
单位: 毫米



OVERALL 外形尺寸 VALUE

Overall Width 总宽	620 mm [△]
Overall Length 总长	520 mm [△]
Overall Height 总高	430 mm [△]
Dry Weight 干重	approx. 48 kg [△]

MOUNTING 安装接口 VALUE

Mounting Hole Pitch 安装孔间距	104 × 118 mm [△]
Mounting Holes 安装孔	M10 × 4 [△]
Thread Depth 螺纹深度	≥ 20 mm [△]

The opposed layout keeps overall height low while extending width laterally — the defining packaging characteristic of the platform.

水平对置结构使发动机在高度方向保持较低, 同时通过横向布局形成明显的 Boxer 结构特征。

[△] Reference dimensions for layout study. 图中所有尺寸为参考尺寸, 具体以实物及生产图纸为准。Mounting point coordinates and interface dimensions must be machined to production engineering drawings. 安装点坐标及接口尺寸应按照生产工程图进行加工。

GENERAL ARRANGEMENT — FRONT, LEFT, TOP, RIGHT VIEWS AND MOUNTING HOLE PATTERN

外形图 · 正视图、左视图、俯视图、右视图及安装孔位



ENGINE INSTALLED IN A MOTORCYCLE PLATFORM — APPLICATION CONTEXT ONLY

发动机装车状态 · 仅作整车集成参考

CB250B uses a modular installation structure suited to motorcycle platform development. This page addresses engine integration only; no vehicle specification is presented.

本页仅说明发动机集成接口，不涉及整车技术参数。

OEM INTERFACES 整车匹配接口

Front Engine Mount 前安装点	M10 [△]
Rear Engine Mount 后安装点	M10 [△]
Crankshaft Output 曲轴输出接口	Per drawing
Clutch Interface 离合器接口	Per drawing
Gearbox Interface 变速箱接口	Per drawing
Intake Interface 进气接口	Airbox
Exhaust Interface 排气接口	Manifold flange
Coolant Interface 冷却液接口	Radiator lines
ECU Interface ECU 接口	Connector
Harness Interface 线束接口	Connector

APPLICATIONS 应用领域

- Street 街车
- Retro Street 复古街车
- Cruiser 巡航摩托车
- Lightweight Performance 轻量化性能车型
- Concept 概念车型
- Custom 定制车型
- Hybrid Platform 混合动力平台

14 PERFORMANCE & DURABILITY

动力性能与耐久性

PROPOSED CALIBRATION

建议标定

MAXIMUM POWER 最大功率

25.0 kW

@ 10,000 rpm [△]

MAXIMUM TORQUE 最大扭矩

28.0 N·m

@ 7,500 rpm [△]

MAXIMUM SPEED 最高转速

11,000 rpm

Permissible limit [△]

IDLE SPEED 怠速

1,500 rpm

± 100 rpm [△]

IMPORTANT — SCOPE OF OUTPUT DATA 重要说明 · 输出数据归属

The 62 hp / 100 N·m figure reported publicly for the P51 is the combined output of the complete hybrid system — the 250cc boxer engine together with its electric motor. It is **not** the output of the CB250B internal combustion engine alone.

P51 公开资料所称的 62 hp / 100 N · m 为 250cc 水平对置发动机与电机组成的完整混合动力系统综合输出，而非 CB250B 内燃机本体输出。

Thermal Stability

热稳定性

Liquid cooling maintains a stable operating temperature band.

Lubrication Reliability

润滑可靠性

Pressure feed sustains oil supply to critical moving parts.

High-RPM Durability

高转速耐久性

High-strength materials and precision machining in the rotating assembly.

Structural Rigidity

结构可靠性

Rigid barrel, head and crankcase structure for long-term operation.

[△] Proposed product calibration for catalog completion. No measured power or torque curve is published in this document. 建议产品标定值。本文件不提供实测功率／扭矩曲线。

15

TECHNICAL DATA SHEET

技术数据总表

CB250B

250CC BOXER TWIN

ENGINE CONSTRUCTION 发动机结构	CB250B
Engine Model 发动机型号	CB250B
Engine Type 发动机类型	Horizontally opposed twin
Cylinder Angle 结构角度	180°
Operating Cycle 工作方式	Four-stroke
Actual Displacement 实际排量	249cc class △
Bore × Stroke 缸径 × 行程	53.5 × 55.2 mm △
Compression Ratio 压缩比	11.5 : 1 ▲
Valvetrain 配气机构	DOHC
Number of Valves 气门数量	8
Cooling 冷却方式	Liquid-cooled
Induction 进气方式	Naturally aspirated
Fuel System 供油方式	EFI
Throttle Body 节气门	34 mm △
Fuel Pressure 燃油压力	3.0 bar △
Starting 启动方式	Electric start
Final Drive Adaptation 最终传动适配	Chain drive
Overall L × W × H 外形尺寸	520 × 620 × 430 mm △
Dry Weight 干重	approx. 48 kg △

PERFORMANCE & SERVICE 性能与维护	CB250B
Maximum Power 最大功率	25.0 kW / 10,000 rpm △
Maximum Torque 最大扭矩	28.0 N·m / 7,500 rpm △
Maximum Engine Speed 最高转速	11,000 rpm △
Idle Speed 怠速	1,500 ± 100 rpm △
Fuel Type 燃料类型	Unleaded, RON 92+ △
Engine Oil 推荐机油	10W-40 △
Oil Grade 机油等级	API SN / JASO MA2 △
Oil Capacity 机油容量	1.8 L △
Coolant Capacity 冷却液容量	approx. 1.6 L △
Spark Plug Gap 火花塞间隙	0.7–0.8 mm △

MAINTENANCE INTERVALS 维护周期	RECOMMENDED △
First Oil Change 首次换油	1,000 km
Oil Change Interval 常规换油周期	5,000 km
Air Filter Inspection 空气滤芯检查	5,000 km
Spark Plug Inspection 火花塞检查	10,000 km
Cooling System Check 冷却系统检查	10,000 km
Coolant Replacement 冷却液更换	20,000 km
Valve Clearance Check 气门间隙检查	20,000 km

▲ Publicly reported figure, not factory -certified. 公开报道数据，非工厂认证参数。 △ Proposed engineering value for catalog completion, pending factory confirmation. 建议工程值，最终以生产图纸及工厂确认数据为准。

CB250B

250CC HORIZONTALLY OPPOSED TWIN-CYLINDER ENGINE

250cc 水平对置双缸发动机

ZHEJIANG CHANGLING BENJIAN MOTORCYCLE CO., LTD.

浙江长铃奔健机车有限公司



BRAND CONTEXT 品牌关联

This document is an engine product presentation. Values marked ▲ are publicly reported figures; values marked △ are proposed engineering values supplied for catalog completion and are not factory-certified specifications. Final specifications, dimensions and interface data are governed by production engineering drawings and written factory confirmation.

本文件为发动机产品说明。标注 ▲ 的为公开报道数据，标注 △ 的为用于目录完整性的建议工程值，均非工厂认证参数。最终技术参数、尺寸及接口数据以生产工程图纸及工厂书面确认为准。技术参数如有变更，恕不另行通知。